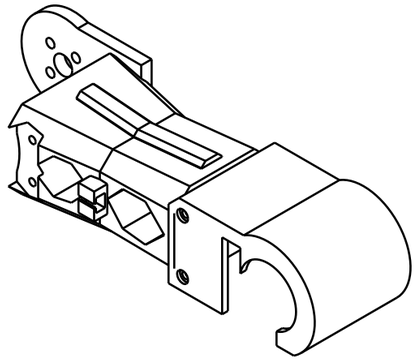
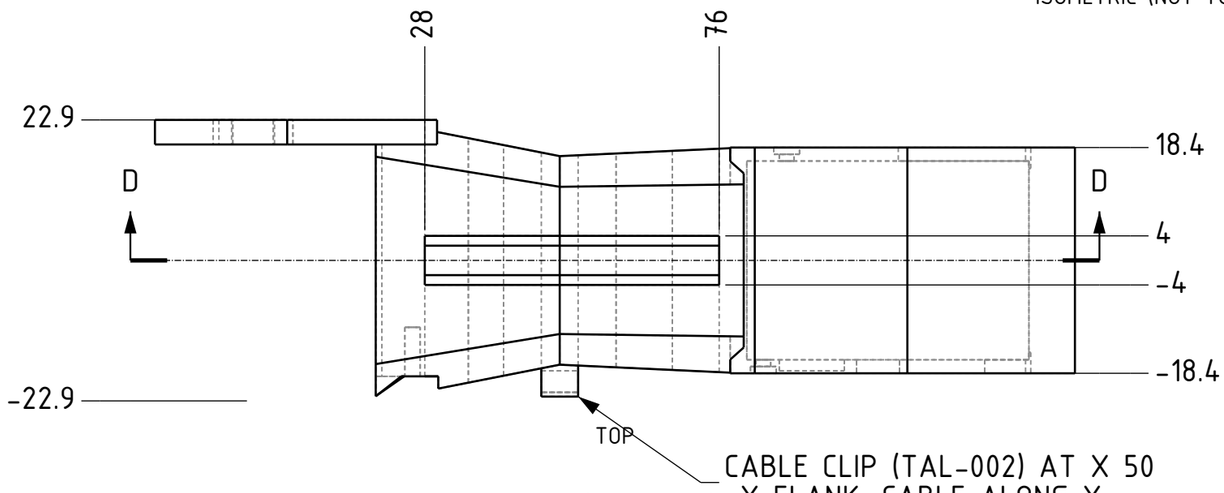
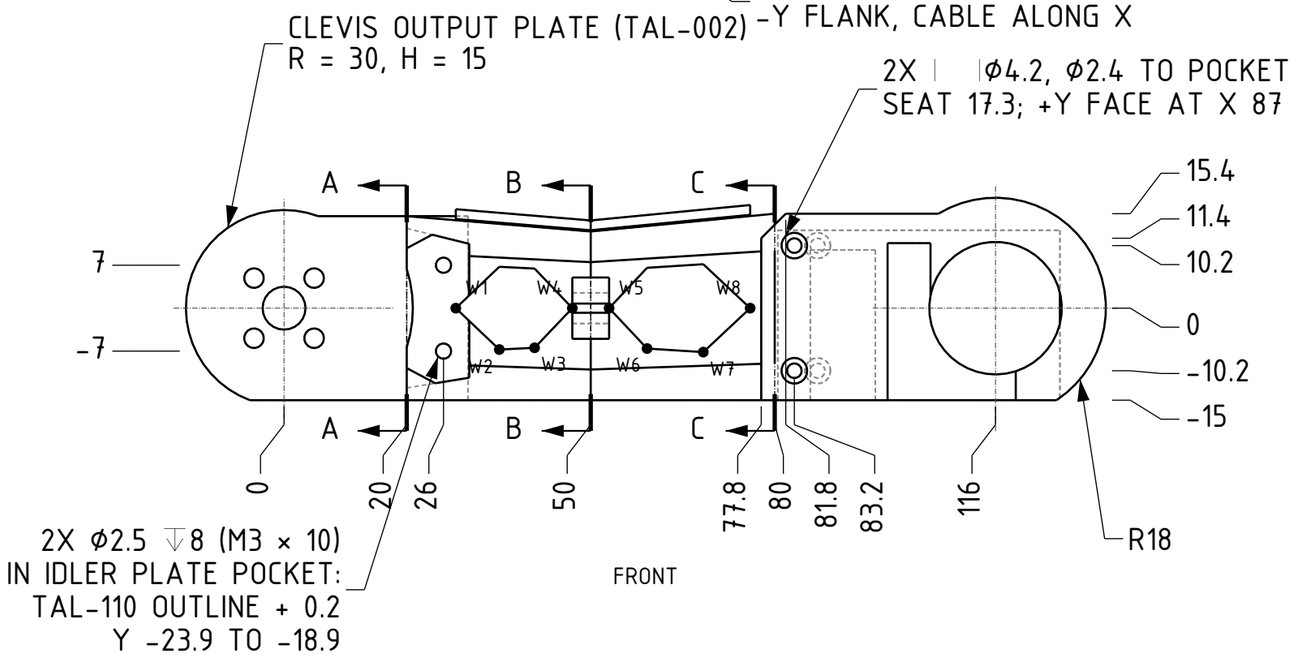


ISOMETRIC (NOT TO SCALE)





WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	35.1	±6.75
W3	40.85	±6.46
W4	47	0
W5	53	0
W6	59.19	±6.58
W7	68.35	±7.16
W8	76	0

FEATURES (in order)

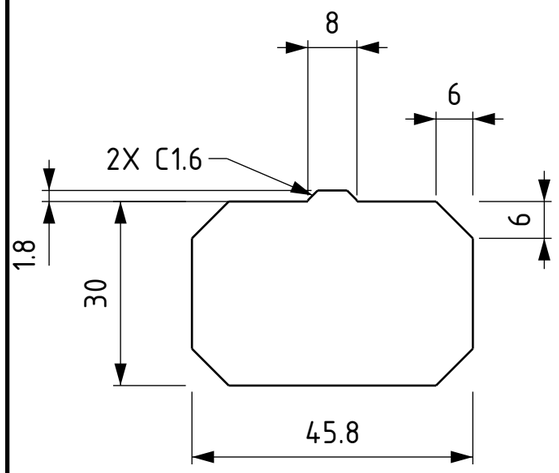
1. Clevis output plate only (TAL-002, +Y plate), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 50, 80. Every profile sits on z −15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (77.8, −15) (116, −15) (116, 15.4) (81.8, 15.4) (77.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 116.
4. Flat underside: cut everything below z −15 (trims the clevis hub and the R18 cylinder).
5. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 76.
6. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
7. Cable clip (TAL-002) on the −Y flank at x 50, cable along X.
8. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (116, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along −Z.
10. Servo entry from below: cut box x 80.5 to 126.5, y ±16.2, z below 12.7 (opens the floor under the whole servo). In the −Y wall: back pad channel x 85.8 to 96.4, y −18 to −16.2, z below −9.5; notch through the wall x 98.4 to 119.3, z below 0 (joins the plug window and the idler hole).
11. Case screw holes (TAL-002, seat 17.3): +Y face at x 87, z ±10.2; −Y face at x 83.2, z ±10.2.
12. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.2, cut from y −23.9 to −18.9.
13. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y −18.9 8 deep (M3 × 10).

NOTES

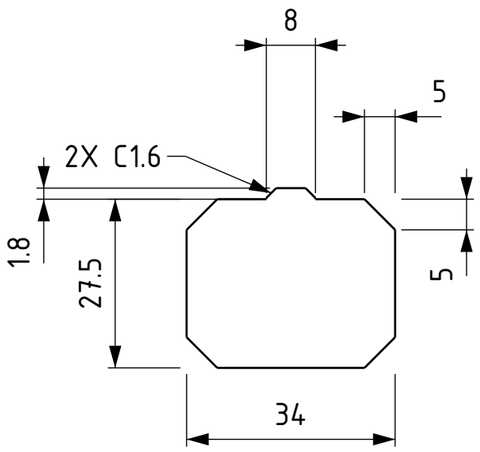
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/−0.2 unless noted; servo pocket clearances are already included.
- Fit the elbow servo from below, shaft to +Y, before its horns and plugs; then 4X M2 × 5 case screws.
- The elbow servo's bus plugs come out through the −Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z −15.

THIRD ANGLE PROJECTION

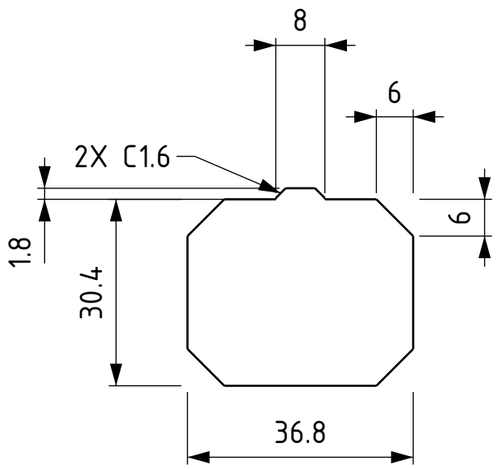
WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 47 g
PART Upper arm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 10 of 21



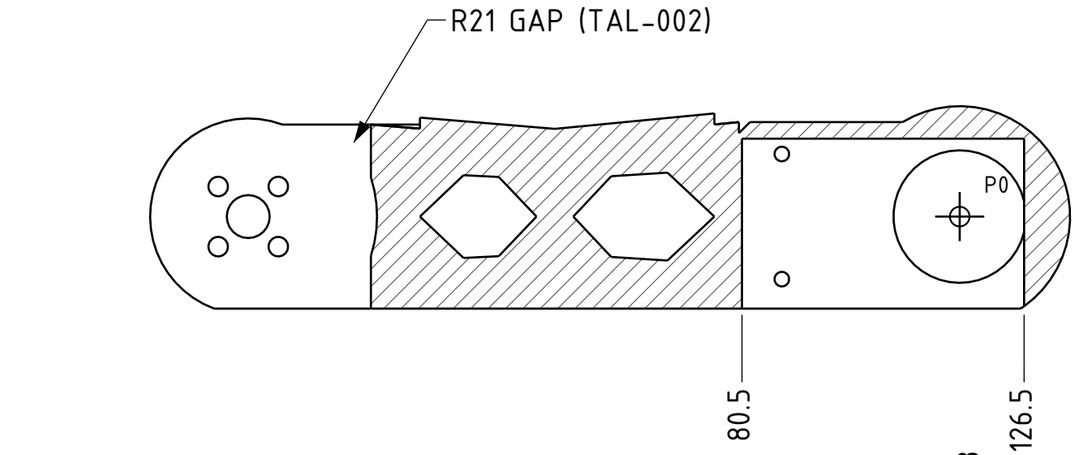
PROFILE A–A (X 20)

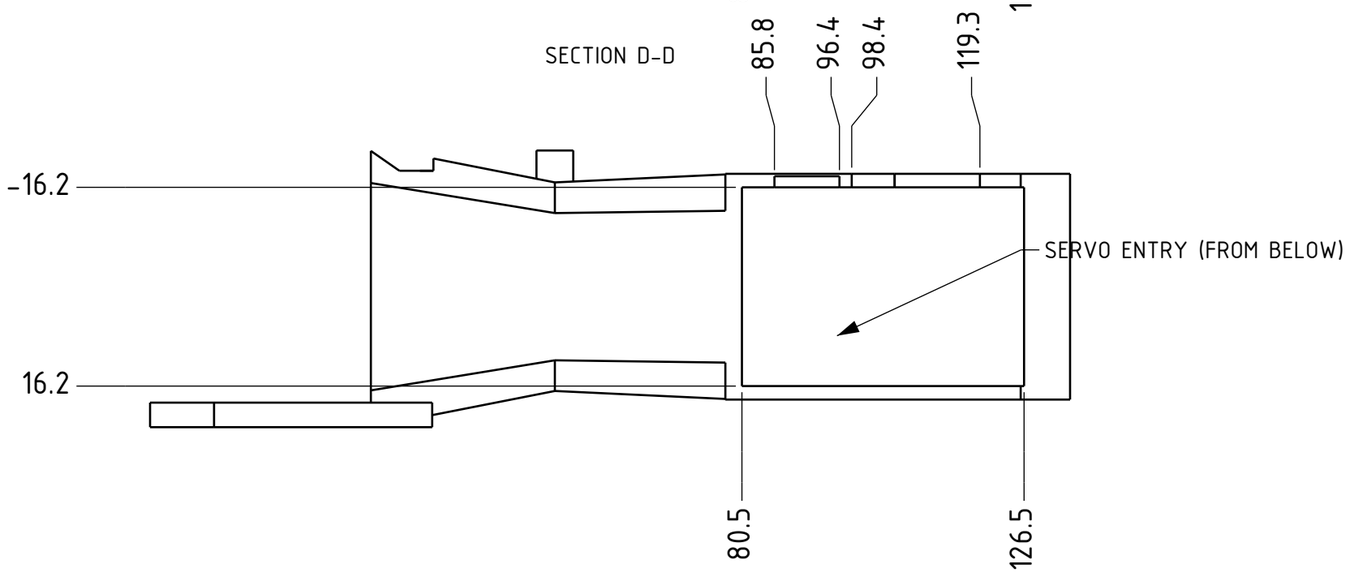


PROFILE B–B (X 50)



PROFILE C–C (X 80)





BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam octagon (corners chamfered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Upper arm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-06
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 11 of 21